Week 3 Part E — written justification

Panel 1: Systolic BP distribution — histogram chosen after trying multiple bin widths; 18 bins show the two subpop

Panel 2: Length of stay by ward — log-scaled box plot because LOS is highly right-skewed; wards are ordered by m

Panel 3: Systolic vs diastolic scatter — the most meaningful clinical pair is the blood-pressure coupling seen in the

Panel 4: Correlation heatmap — diverging scale from -1 to 1, masked upper triangle, annotations, and excluded de

Every parameter was declared: bins = 18, log-scale LOS, 3,000-point subsample for the scatter, and synthetic-data

Cleaning rules: ward spellings standardized to four names; dates parsed from three formats; BMI sentinel 999 repla

Ethics clause: readings above 250 bpm were removed as device faults (n = 37). Mean heart rate moves from 78.8

The plotting choice follows Week 2 design principles: colour-blind-safe palette, no dual-axis chart, direct labels whe